

Ngoc Thien Kim Nguyen

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EDUCATION

Julius-Maximilians University of Würzburg

Master of Science: Biosciences – Molecular and Computational Biology. GPA 1.7

International University, Vietnam National University

Bachelor of Science: Biotechnology. GPA 1.8

Würzburg, Germany

Oct. 2023 – Now

Ho Chi Minh City, Vietnam

Oct. 2017 – Apr. 2022

RESEARCH PROJECTS

Theodor Boveri Institute, Julius-Maximilians-Universität Würzburg

Master's Thesis — Grade: 1.3/4.0

Würzburg, Germany

Mar. 2025 – Jun. 2026

Title: Characterisation of the RNA Binding Behaviour of the **RNA-Binding Protein MATRN3** in Murine Cells

- Developed and optimized **computational workflow for transcriptome-wide RNA-editing data analysis** to characterize genuine MATRN3–RNA interactions.
- Integrated multi-dataset **HyperTRIBE, iCLIP and bulk RNA-seq** datasets to identify reproducible and biologically meaningful targets of MATRN3-RBP with high confidence.
- Performed **feature selection and statistical modelling (R)**, differential gene expression analysis **DESeq2**.
- Examine sequencing quality, inspect SNPs and genomic visualization using **IGV** from BAM, BigWig, Bed files.
- Analyzed alternative splicing using a **Nextflow/MAJIQ** pipeline.

Computational RNA Biology Intern

Mar. 2025 – Sep. 2025

Project: Regulatory Impact of 3' Splice-Site Factors on Gene Expression

- Investigated the effects of **U2AFs and SF1** on gene expression and alternative splicing using **ENCODE RNA-seq datasets** from HepG2 and K562 cells.
- Processed RNA-seq data using **FastQC, MultiQC, Trimmomatic, STAR, HTSeq-count, featureCounts** in a **Linux/HPC environment**.
- Performed **differential expression analysis (DESeq2), clustering, and GO enrichment**, together with **alternative-splicing analysis (MAJIQ/VOILA)** to characterize transcriptional and splicing responses to factor perturbation.
- Investigated **regulatory relationships within the splice-site factor network** via clustering and correlation analysis.
- Inspected sequencing quality, genomic regions and splicing sites using **IGV**.

ADDITIONAL WORKING EXPERIENCE

Bioinformatics Department, Julius-Maximilians-Universität Würzburg

System Biology research

Würzburg, Germany

Oct. 2023 – Dec. 2023

- Constructed and analyzed metabolic networks of *Candida albicans* using **YANAsquare** and **METATOOL**.
- Curated pathway data from multi-databases (**KEGG, Rhea, NCBI, UniProt**) and literature review.
- Performed **Elementary Flux Mode** analysis and interpreted **Metabolic flux distributions** across fungal metabolic networks.

Cell Biotechnology Laboratory, International University

Bachelor's Thesis

Ho Chi Minh City, Vietnam

Apr. 2022 – Oct. 2022

- Optimised polyphenol extraction from *Perilla frutescens* using statistical method **Response Surface Methodology (RSM)**.
- Assessed antioxidant activity using Folin–Ciocalteu and FRAP assays and evaluated DNA-protection effects in *S. cerevisiae* and *E. coli* models.

The Applied Biotechnology Institute, Molecular Department

Bioinformatics Intern

Ho Chi Minh City, Vietnam

Aug. 2021 – Dec. 2021

- Performed molecular identification of fungal isolates using **Sanger sequencing and ITS-based sequence analysis**.
- Edited and quality-controlled sequencing chromatograms using **ATGC**, followed by sequence alignment and **BLAST analysis against NCBI GenBank** for species identification.
- Built and curated a reference database of **~200 fungal species**, integrating sequence identifications, GenBank accession information, literature references and matching images for species comparison.

EXTRACURRICULAR ACTIVITIES

Vietnam National University Academic Newsletter

Ho Chi Minh City, Vietnam

Research Assistance

2021

- Analyzed and summarized **scientific publications** to extract key insights for non-specialist audiences.
- Translated clinical reports and biomedical research on **COVID-19 therapies** and vaccines into **Vietnamese**, improving accessibility and knowledge dissemination.
- Publication: *"Treatments for COVID-19 in the World", "Vaccine Efficacy Against SARS-CoV-2 Virus Variants and Related Questions When Vaccinating Patients Previously Infected with COVID-19"*.

Third prize in Vietnam's Amazing Student Contest–RMIT University Ho Chi Minh City, Vietnam

- Project: Anti-microplastic pollution campaign.

2019

TECHNICAL SKILLS & RESEARCH INTERESTS

Programming: R, Python, Shell/Bash

Data Analysis: Bioconductor, DESeq2, GenomicRanges, MAJIQ, IGV, BLAST

Infrastructure: Git, Linux, HPC/Slurm, Conda,

Research Interests: Multi-omics, RNA Biology, Alternative Splicing, Data mining, Systems Biology

Languages: Vietnamese (Native), English (Proficiency), German (B1)